

Enterprise Network Simulation

Cisco Packet Tracer — Full Design & Configuration

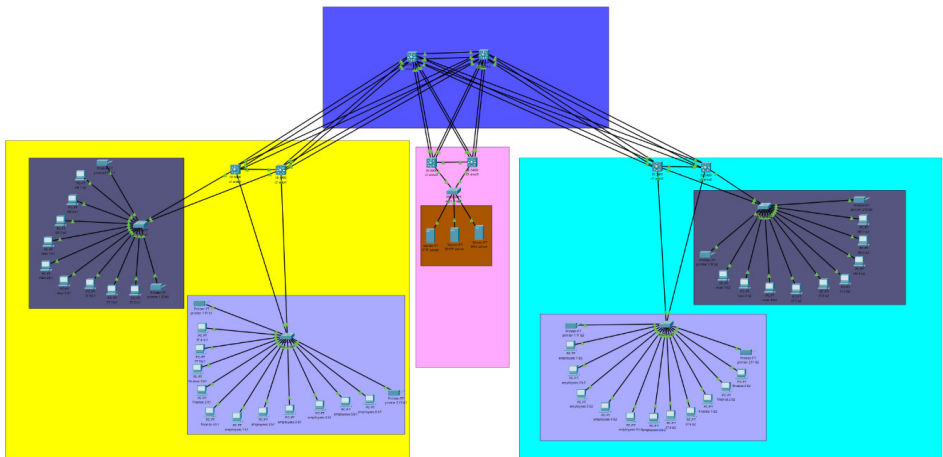


Figure 1: Full Network Topology — 4-Area OSPF Enterprise Design

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Level	CCNA Preparation
Tool	Cisco Packet Tracer
Duration	~3 Days
Status	v1.0 — Functional (NAT / ACL / Switch Security planned for v2)

1. Project Overview

This project simulates a real-world enterprise network built entirely from scratch in Cisco Packet Tracer. The design covers a multi-building company with centralized services, full redundancy, load balancing, and fault tolerance at every layer of the network stack.

The network is divided into 4 OSPF areas: a backbone core (Area 0), two company buildings (Areas 1 and 2), and a dedicated server farm (Area 3). Every design decision was made to reflect real enterprise practices — not just to make the lab work.

Technologies Used

Category	Technologies
Routing	OSPF Multi-Area, Route Summarization, Point-to-Point Links
Redundancy	HSRP, Dual Uplinks, EtherChannel (LACP)
Layer 2	VLANs, Rapid PVST+, STP Load Balancing, Trunk Links
STP Security	PortFast, BPDUGuard, RootGuard
Services	DHCP (centralized + relay), DNS, TFTP
Switching	Layer 3 SVIs for Inter-VLAN Routing

2. Area 0 — Backbone Core

Area 0 is the OSPF backbone. All other areas connect to it and routing between areas passes through this core. Three Layer 3 switches form the backbone, interconnected to provide a non-blocking core fabric.

Feature	Detail
Devices	3x Layer 3 Switches
Protocol	OSPF Area 0
Uplinks	Dual interfaces per area switch
Bundling	EtherChannel (LACP) on all uplinks
Benefit	Redundancy + load balancing + increased bandwidth

Each area connects to the backbone via two physical interfaces bundled into a single EtherChannel. This eliminates STP blocking on redundant links while doubling the available bandwidth. If one physical interface fails, traffic continues on the remaining member without any network interruption.

3. Area 1 & Area 2 — Company Buildings

Areas 1 and 2 represent two physical buildings of the same company. Each building has its own IP address range and hosts all company departments. The designs are identical — only the IP ranges differ.

IP Addressing

Area	Building	IP Range	Purpose
Area 1	Building 1	192.168.0.0/16	All VLANs subneted from this range
Area 2	Building 2	172.31.0.0/16	All VLANs subneted from this range

VLAN Structure

VLAN ID	Name	Area 1 Subnet	Area 2 Subnet
10	HR	192.168.10.0/24	172.31.10.0/24
20	IT	192.168.20.0/24	172.31.20.0/24
30	Management	192.168.30.0/24	172.31.30.0/24
40	Finance	192.168.40.0/24	172.31.40.0/24
50	Employees	192.168.50.0/24	172.31.50.0/24
60	Printers	192.168.60.0/24	172.31.60.0/24
200	Native VLAN	—	—

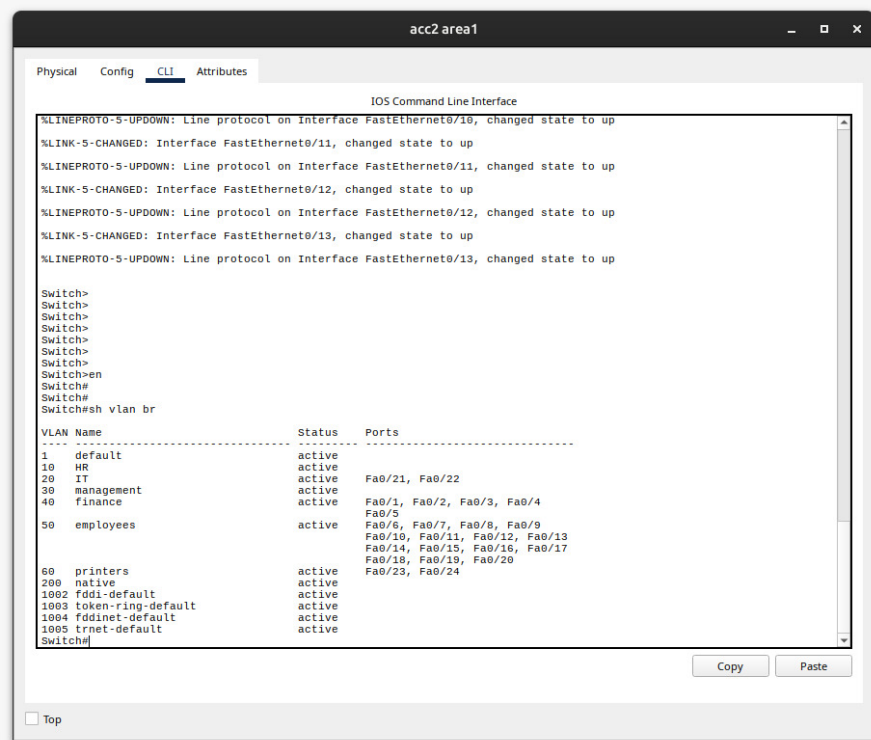


Figure 2: VLAN brief on Access Switch — Area 1

3.1 STP Load Balancing — Rapid PVST+

Instead of having one switch carry all traffic, STP is configured to distribute VLANs evenly between C1 and C2. This means both switches actively forward traffic at all times — no idle standby links.

Switch	Root Bridge For	Secondary For
C1	VLANs 10, 20, 30, 200	VLANs 40, 50, 60, 1
C2	VLANs 40, 50, 60, 1	VLANs 10, 20, 30, 200

Commands used:

```
! On C1 spanning-tree vlan 10,20,30,200 root primary spanning-tree vlan 40,50,60,1 root secondary
! On C2 spanning-tree vlan 40,50,60,1 root primary spanning-tree vlan 10,20,30,200 root secondary
```

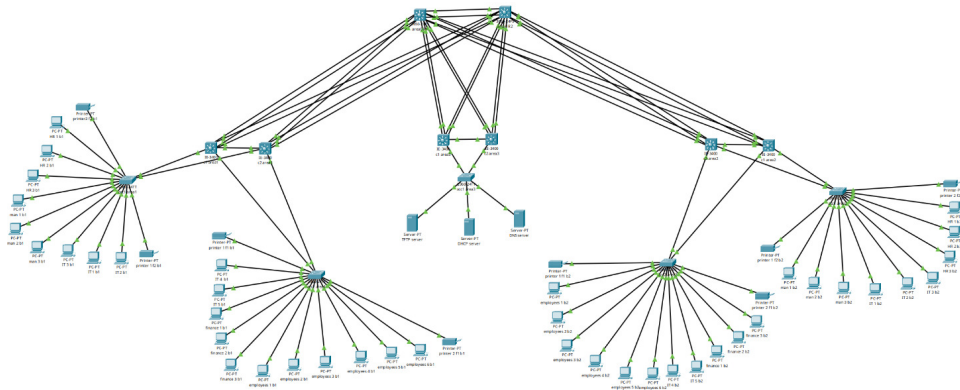


Figure 3: STP Verification — C1 Area 1 is Root for VLAN 10, 20, 30

3.2 STP Security Features

Feature	Applied On	Purpose
Rapid PVST+	All switches	Convergence in ~6 seconds instead of 30-50
PortFast	End-device ports	Skip STP states — instant connectivity
BPDUGuard	End-device ports	Shutdown port if unexpected BPDU received
RootGuard	Core to Access uplinks	Prevent unauthorized root bridge election

3.3 HSRP — Gateway Redundancy & Load Balancing

HSRP provides a Virtual IP (VIP) for each VLAN that end devices use as their default gateway. If the active router fails, the standby takes over automatically. The configuration mirrors the STP split — C1 is active for VLANs 10/20/30, C2 is active for VLANs 40/50/60 — so both switches handle real traffic simultaneously.

VLAN	Active (Priority 110)	Standby (Priority 90)	Virtual IP
10 — HR	C1	C2	192.168.10.1
20 — IT	C1	C2	192.168.20.1
30 — Mgmt	C1	C2	192.168.30.1
40 — Finance	C2	C1	192.168.40.1
50 — Employees	C2	C1	192.168.50.1
60 — Printers	C2	C1	192.168.60.1

Key command:

```
! C1 – VLAN 10 (Active) interface Vlan10 standby 1 ip 192.168.10.1 standby 1 priority 110
standby 1 preempt ! C2 – VLAN 10 (Standby) interface Vlan10 standby 1 ip 192.168.10.1
standby 1 priority 90
```

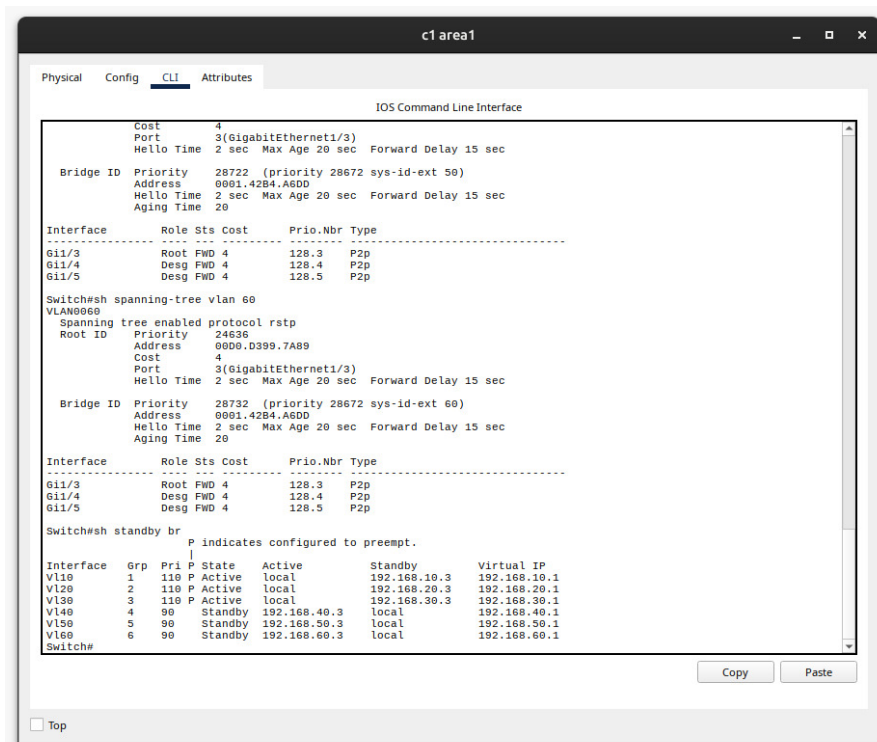


Figure 4: HSRP Verification — C1 Active on VLANs 10/20/30, C2 Active on 40/50/60

3.4 OSPF Route Summarization

Each ABR (Area Border Router) summarizes its area's entire IP range into a single route before advertising to Area 0. This keeps the backbone routing table clean and reduces OSPF update overhead when subnets change inside an area.

Area	Summary Route	Command
Area 1	192.168.0.0/16	area 1 range 192.168.0.0 255.255.0.0
Area 2	172.31.0.0/16	area 2 range 172.31.0.0 255.255.0.0
Area 3	10.10.100.0/24	area 3 range 10.10.100.0 255.255.255.0

```
Switch>
Switch#sh ip ro
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 21 subnets, 3 masks
C    10.0.0.0/30 is directly connected, Port-channel20
L    10.0.0.2/32 is directly connected, Port-channel20
O    10.0.1.0/30 [110/2] via 10.0.1.10, 00:16:12, Port-channel5
L    10.0.1.2/32 is directly connected, Port-channel20
O    10.0.1.4/30 [110/2] via 10.0.1.13, 00:16:12, Port-channel6
L    10.0.1.6/30 [110/2] via 10.0.0.1, 00:16:12, Port-channel20
C    10.0.1.8/30 is directly connected, Port-channel5
L    10.0.1.9/32 is directly connected, Port-channel5
C    10.0.1.12/30 is directly connected, Port-channel6
L    10.0.1.14/32 is directly connected, Port-channel6
O    10.0.2.0/30 [110/2] via 10.0.2.10, 00:16:12, Port-channel7
L    10.0.2.2/30 [110/2] via 10.0.0.1, 00:16:12, Port-channel20
O    10.0.2.4/30 [110/2] via 10.0.2.14, 00:16:12, Port-channel8
L    10.0.2.6/30 [110/2] via 10.0.0.1, 00:16:12, Port-channel20
C    10.0.2.8/30 is directly connected, Port-channel7
L    10.0.2.9/32 is directly connected, Port-channel7
C    10.0.2.12/30 is directly connected, Port-channel8
L    10.0.2.13/32 is directly connected, Port-channel8
O    10.0.3.0/30 [110/2] via 10.0.3.10, 00:16:12, Port-channel11
L    10.0.3.2/30 [110/2] via 10.0.0.1, 00:16:12, Port-channel20
O    10.0.3.4/30 [110/2] via 10.0.3.14, 00:16:12, Port-channel12
L    10.0.3.6/30 [110/2] via 10.0.0.1, 00:16:12, Port-channel20
C    10.0.3.8/30 is directly connected, Port-channel11
L    10.0.3.9/32 is directly connected, Port-channel11
C    10.0.3.12/30 is directly connected, Port-channel12
L    10.0.3.13/32 is directly connected, Port-channel12
O IA 10.10.0.0/24 [110/2] via 10.0.3.14, 00:16:12, Port-channel12
O IA 172.31.0.0/16 [110/2] via 10.0.3.10, 00:16:12, Port-channel11
O IA 192.168.0.0/16 [110/2] via 10.0.2.14, 00:16:12, Port-channel8
O IA 192.168.0.0/16 [110/2] via 10.0.1.13, 00:16:12, Port-channel6

Switch#
```

Figure 5: OSPF Routing Table on Area 0 R1 — showing O IA summary routes

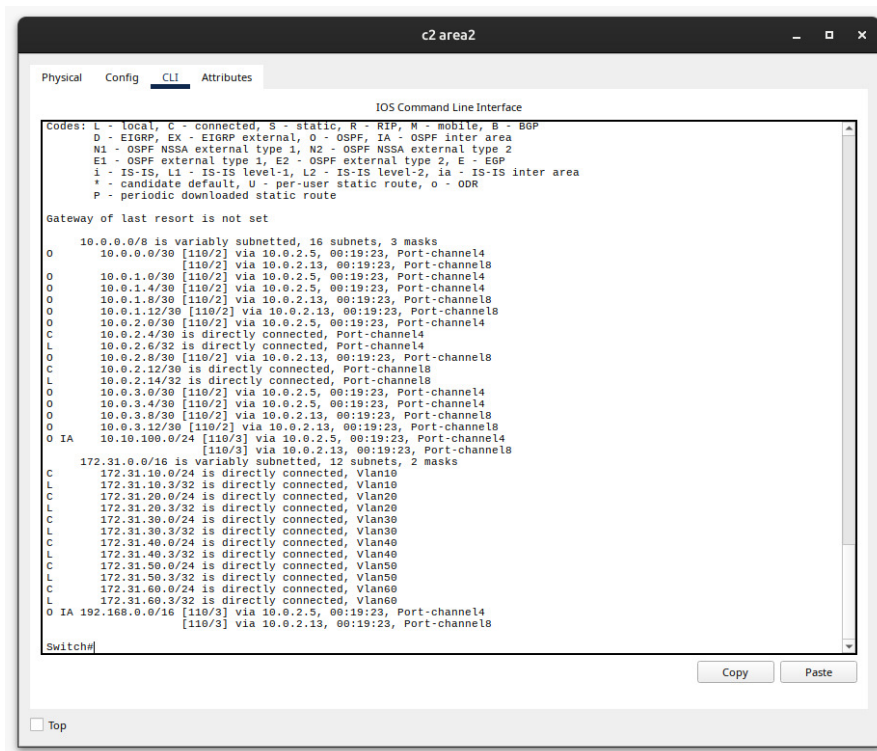


Figure 6: OSPF Routing Table on Area 0 R2 — ECMP via Port-channels

4. Area 3 — Server Farm

Area 3 is a dedicated, isolated area hosting all centralized services. It uses a simple design with a single server VLAN and two core switches for redundancy.

Server	IP	Role
DHCP Server	10.10.100.10	Assigns IPs to all VLANs in Areas 1 & 2
DNS Server	10.10.100.20	Resolves hostnames for all network devices
TFTP Server	10.10.100.30	Config backup and IOS image storage

Feature	Detail
VLANs	VLAN 100 (Server), VLAN 200 (Native)
HSRP VIP	10.10.100.1
STP	C1 = Root Bridge, C2 = Secondary
Security	PortFast + BPDUGuard + RootGuard
Server IPs	All static — registered in DNS

5. DHCP — Centralized Configuration

All DHCP pools are hosted on the central DHCP server in Area 3. The `ip helper-address` command on each SVI forwards client DHCP broadcasts across area boundaries to the server. The default gateway in each pool is set to the HSRP Virtual IP — ensuring clients always reach a working gateway even if one core switch fails.

Key `ip helper-address` command:

```
interface Vlan10 ip address 192.168.10.2 255.255.255.0 ip helper-address 10.10.100.10
standby 1 ip 192.168.10.1
```

DHCP Pool Summary:

Pool Name	Gateway (VIP)	DNS	Start IP	Max Hosts
VLAN 10 Area 1	192.168.10.1	10.10.100.20	192.168.10.9	100
VLAN 20 Area 1	192.168.20.1	10.10.100.20	192.168.20.9	100
VLAN 30 Area 1	192.168.30.1	10.10.100.20	192.168.30.9	100
VLAN 40 Area 1	192.168.40.1	10.10.100.20	192.168.40.9	100
VLAN 50 Area 1	192.168.50.1	10.10.100.20	192.168.50.9	100
VLAN 10 Area 2	172.31.10.1	10.10.100.20	172.31.10.9	100
VLAN 20 Area 2	172.31.20.1	10.10.100.20	172.31.20.9	100
VLAN 30 Area 2	172.31.30.1	10.10.100.20	172.31.30.9	100
VLAN 40 Area 2	172.31.40.1	10.10.100.20	172.31.40.9	100
VLAN 50 Area 2	172.31.50.1	10.10.100.20	172.31.50.9	100

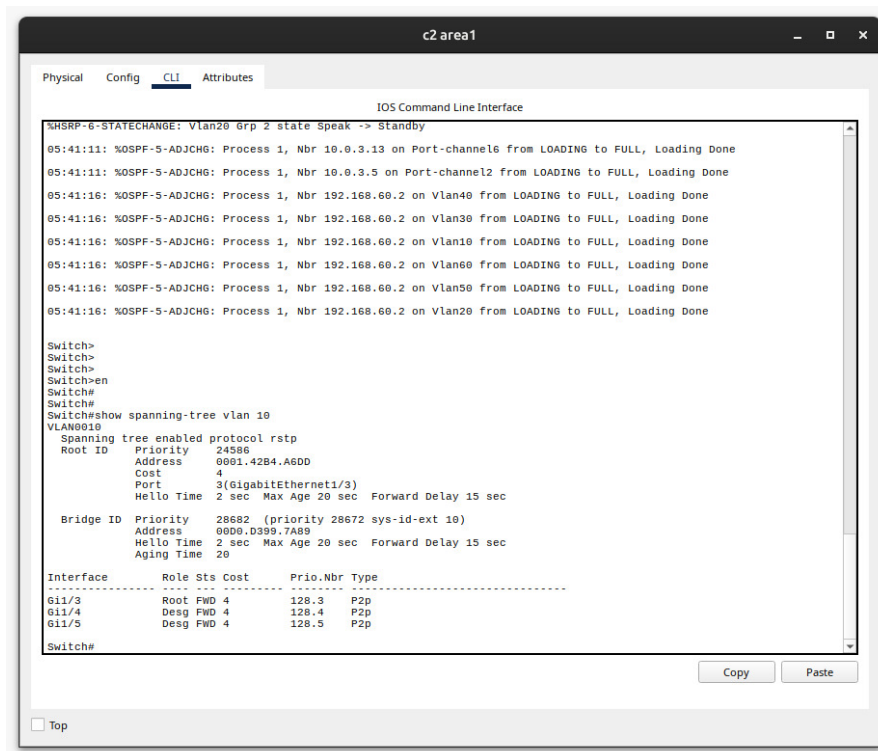


Figure 7: DHCP Server — all pools configured for Areas 1 and 2

6. DNS — Hostname Resolution

All printers (VLAN 60) have static IPs and are registered in the DNS server by hostname. All servers are also registered. This allows any device on the network to reach a printer or server by name — across VLANs and areas — without knowing the IP.

Hostname	IP	Type
dhcp-server	10.10.100.10	A Record
dns-server	10.10.100.20	A Record
tftp-server	10.10.100.30	A Record
printer1-f1-b1	192.168.60.30	A Record
printer1-f1-b2	172.31.60.10	A Record
printer1-f2-b1	192.168.60.10	A Record
printer2-f1-b1	192.168.60.40	A Record
printer2-f2-b2	172.31.60.30	A Record

The screenshot shows the 'DHCP server' configuration window with the 'Services' tab selected. The 'DHCP' service is enabled. The configuration fields include: Interface (FastEthernet0), Pool Name (serverPool), Default Gateway (0.0.0.0), DNS Server (0.0.0.0), Start IP Address (10.10.100.10), Subnet Mask (255.255.255.0), Maximum Number of Users (512), TFTP Server (0.0.0.0), and WLC Address (0.0.0.0). Below these fields is a table listing various DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
vlan 50 area 2	172.31.50.1	10.10.100.20	172.31.50.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 40 area 2	172.31.40.1	10.10.100.20	172.31.40.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 30 area 2	172.31.30.1	10.10.100.20	172.31.30.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 20 area 2	172.31.20.1	10.10.100.20	172.31.20.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 10 area 2	172.31.10.1	10.10.100.20	172.31.10.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 20 area 1	192.168.20.1	10.10.100.20	192.168.20.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 10 area 1	192.168.10.1	10.10.100.20	192.168.10.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 40 area 1	192.168.40.1	10.10.100.20	192.168.40.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 50 area 1	192.168.50.1	10.10.100.20	192.168.50.9	255.255.255.0	100	10.10.100.30	0.0.0.0
vlan 30 area 1	192.168.30.1	10.10.100.20	192.168.30.9	255.255.255.0	100	10.10.100.30	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.10.100.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Figure 8: DNS Server — all printers and servers registered by hostname

7. Verification

7.1 — PC Receives IP via DHCP

HR 1 b1 (VLAN 10, Area 1) successfully received an IP address from the centralized DHCP server in Area 3. The default gateway shown is 192.168.10.1 — the HSRP Virtual IP — confirming the ip helper-address relay is working correctly across areas.

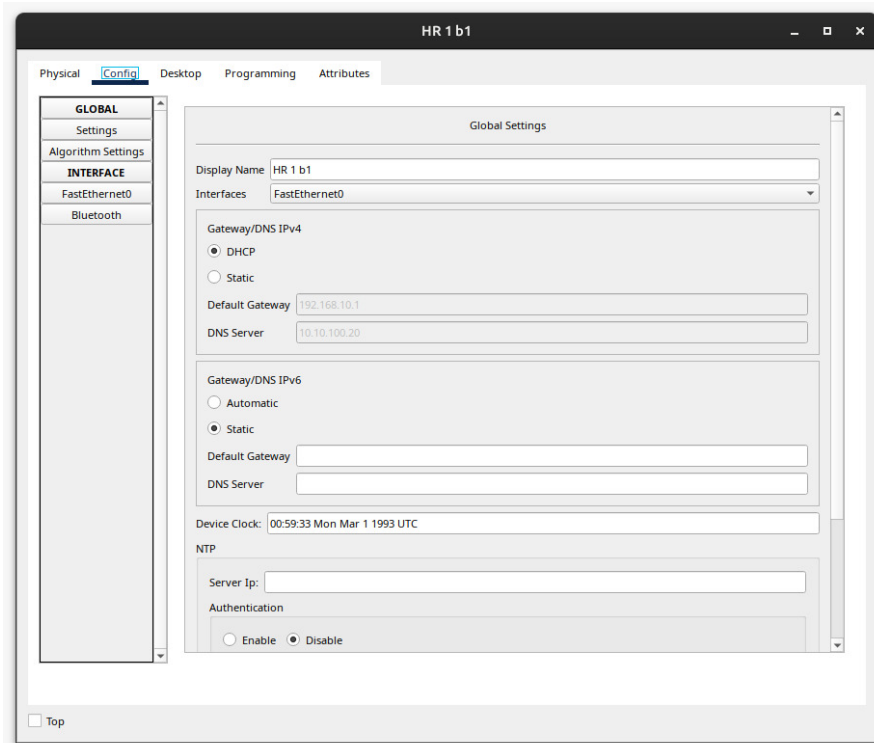


Figure 9: HR 1 b1 — DHCP assigned 192.168.10.11, Gateway 192.168.10.1 (HSRP VIP)

7.2 — Cross-Area Ping by Hostname

A management PC in Area 1 (Building 1) successfully pinged a printer in Area 2 (Building 2) using only its DNS hostname. This confirms: DNS resolution working, inter-VLAN routing working, OSPF inter-area routing working — all end-to-end.

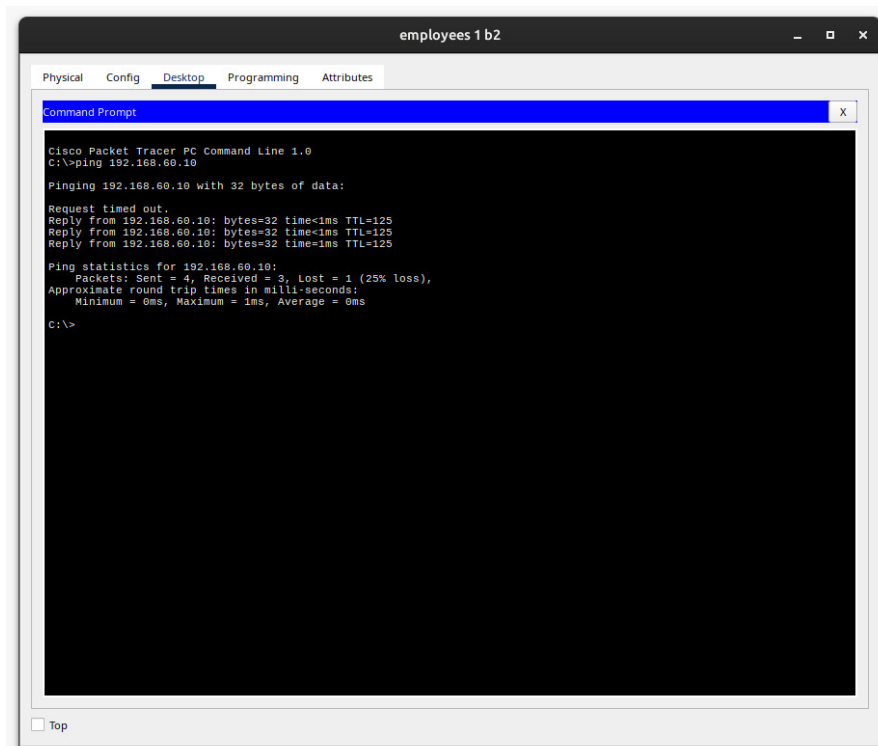


Figure 10: man 1 b1 pings printer1-f2-b2 (172.31.60.30) by hostname — cross-area success

8. Planned Improvements (v2)

The following features were not yet studied at the time this project was built. They will be added in version 2 as skills are acquired:

Feature	Purpose
NAT (Network Address Translation)	Simulate internet access from private address ranges
ACLs (Access Control Lists)	Filter traffic between VLANs and areas
Port Security	Restrict which MAC addresses can connect per port
DHCP Snooping	Prevent rogue DHCP servers on the network
Dynamic ARP Inspection	Protect against ARP poisoning attacks
NTP	Centralized time synchronization across all devices
Syslog	Centralized logging for all network events